

Class 21: Obesity is contagious, maybe

Matthew J. Salganik

Sociology 204: Social Networks
Princeton University

Monday, April 14, 2025

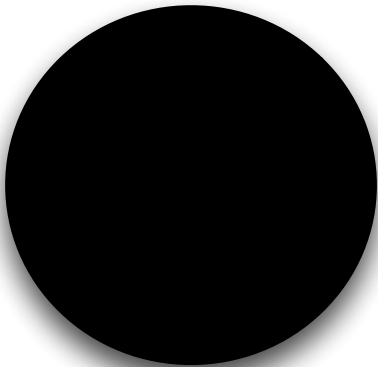


APRIL 2025

SUN	MON	TUE	WED	THU	FRI	SAT
30	31	1	2 Assignment 5 (design) due 11am	3	4 Submit revised design 11:59pm	5 Begin experiment
6	7	8	9 Assignment 6 (progress) due 11am	10	11	12
13	14	15 End experiment	16 Assignment 7 (results) due 11am	17	18	19
20	21	22	23 Assignment 8 (reflection) due 11am	24	25 End of semester	26
27	28	29	30	1	2	3

- ▶ Chunk 1: Personal networks
- ▶ Chunk 2: Social media
- ▶ Chunk 3: Social contagion

1. Thompson, C. (2009). Are your friends making you fat? *New York Times Magazine*.
2. Christakis, N.A. and Fowler, J.H. (2007). The spread of obesity in a large social network over 32 years. *New England Journal of Medicine*.
3. Johns, D. (2011). Disconnected? *Slate*.
4. Lyons, R. (2011). The spread of evidence-poor medicine via flawed social-network analysis. *Statistics, Politics and Policy*.
5. Gelman (2024) Here's a useful response by Christakis to criticisms of the contagion-of-obesity claims *Blog Post*





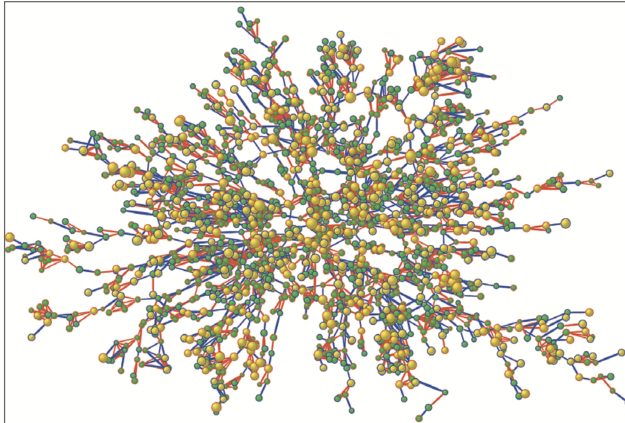


Figure 1. Largest Connected Subcomponent of the Social Network in the Framingham Heart Study in the Year 2000.

Each circle (node) represents one person in the data set. There are 2200 persons in this subcomponent of the social network. Circles with red borders denote women, and circles with blue borders denote men. The size of each circle is proportional to the person's body-mass index. The interior color of the circles indicates the person's obesity status: yellow denotes an obese person (body-mass index, ≥ 30) and green denotes a nonobese person. The colors of the ties between the nodes indicate the relationship between them: purple denotes a friendship or marital tie and orange denotes a familial tie.

Here's what I hope you will take away from class today

- ▶ network-centric view of health
- ▶ description of patterns in the data
- ▶ challenges with making causal claims from that data without an experiment
- ▶ a desire to take more statistics classes
- ▶ excited for the experimental readings for Wednesday

Predictions vs causation

- ▶ If my friend becomes obese should you predict that I will become obese?
- ▶ Does my friend becoming obese cause me to become obese?

The Spread of Obesity in a Large Social Network over 32 Years

Nicholas A. Christakis, M.D., Ph.D., M.P.H., and James H. Fowler, Ph.D.

Glossary

Ego: The person whose behavior is being analyzed.

Alter: A person connected to the ego who may influence the behavior of the ego.

Node: An object that may or may not be connected to other objects in a network. In this study, nodes represent people in the Framingham Heart Study cohorts.

Tie: A connection between two nodes that can be either one-way (directed) or two-way (bilateral). In this study, all family ties (e.g., between siblings and parents) as well as marital ties are bilateral, but friendship ties are directional since a subject may identify someone as a friend who does not necessarily identify that person as a friend in return.

Degree of separation: The social distance between two people as measured by the smallest number of intermediaries between an ego and other members of the network. For a given ego, alters are degree 1, since they are directly connected to the ego. Nodes that are connected to the alters but not to the ego are degree 2 (alters' alters). Nodes that are connected to the alters' alters but not to the ego are degree 3, and so on.

Homophily: The tendency for people to choose relationships with people who have similar attributes.

Induction: The spread of a behavior or trait from one person to another.

Cluster: A group of nodes, each of which is connected to at least one other node in the group.

Connected component: Part of a social network in which all persons have a social tie to at least one other person and no person is connected to a member of any other component of the network.

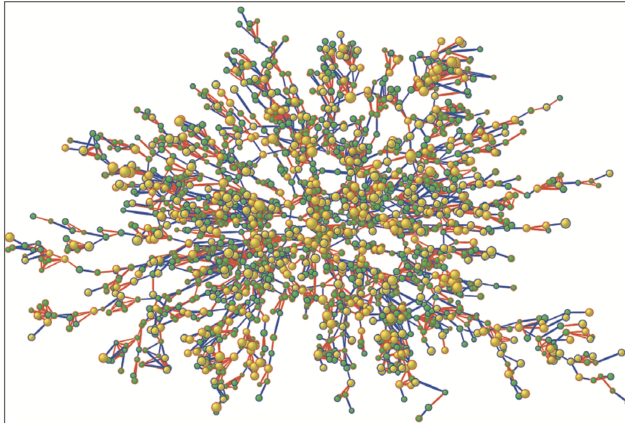


Figure 1. Largest Connected Subcomponent of the Social Network in the Framingham Heart Study in the Year 2000.

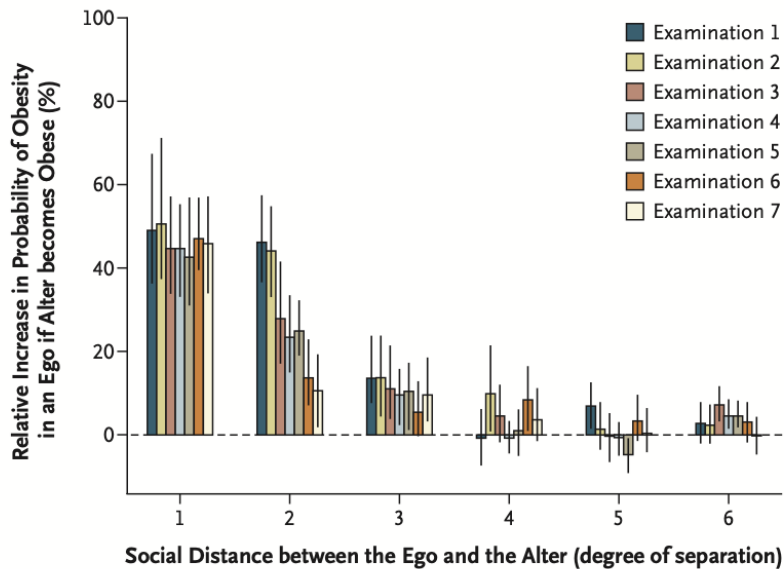
Each circle (node) represents one person in the data set. There are 2200 persons in this subcomponent of the social network. Circles with red borders denote women, and circles with blue borders denote men. The size of each circle is proportional to the person's body-mass index. The interior color of the circles indicates the person's obesity status: yellow denotes an obese person (body-mass index, ≥ 30) and green denotes a nonobese person. The colors of the ties between the nodes indicate the relationship between them: purple denotes a friendship or marital tie and orange denotes a familial tie.

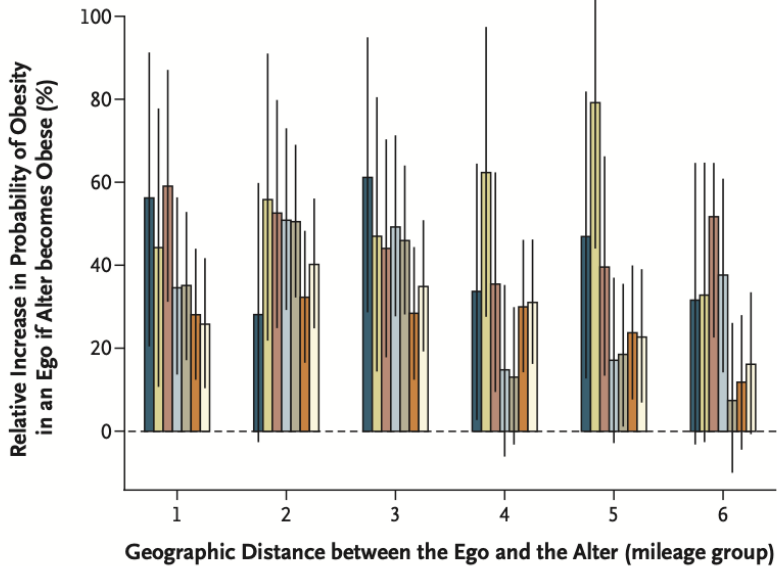
Why might obese people tend to be connected to obese people and non-obese people tend to be connected to non-obese people?

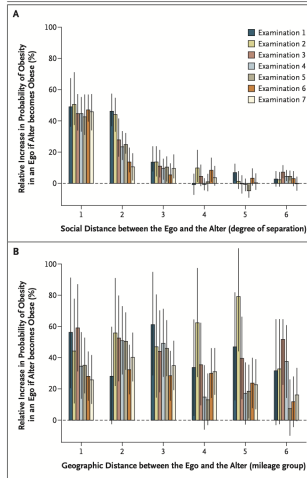
Think - pair- share

We considered three explanations for the clustering of obese people. First, egos might choose to associate with like alters (“homophily”).^{21,23,24} Second, egos and alters might share attributes or jointly experience unobserved contemporaneous events that cause their weight to vary at the same time (confounding). Third, alters might exert social influence or peer effects on egos (“induction”). Distinguishing the interpersonal induction of obesity from homophily requires dynamic, longitudinal network information about the emergence of ties between people (“nodes”) in a network and also about the attributes of nodes (i.e., repeated measures of the body-mass index).²⁵

- ▶ obese people could prefer to be friend with obese people and/or skinny people could prefer to be friend with skinny people (homophily on observed traits)
- ▶ people who both like ice cream could become friends (homophily in unobserved traits)
- ▶ obese people may live on one side of town with lots of McDonalds, and non-obese on the other with no McDonalds (common environment)
- ▶ obesity spreads (contagion, which they call induction)

A

B



Some evidence social distance is more important than physical distance

Three different kinds of studies:

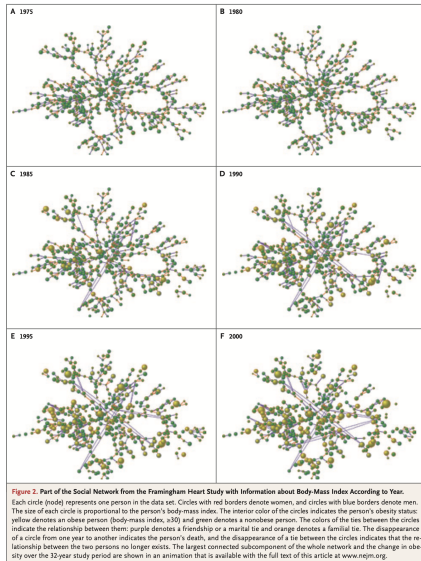
- ▶ cross-sectional study (data from one moment in time): causality very hard

Three different kinds of studies:

- ▶ cross-sectional study (data from one moment in time): causality very hard
- ▶ longitudinal non-experimental study (repeated snapshots with no randomized intervention): causality a bit easier

Three different kinds of studies:

- ▶ cross-sectional study (data from one moment in time): causality very hard
- ▶ longitudinal non-experimental study (repeated snapshots with no randomized intervention): causality a bit easier
- ▶ randomized controlled experiment (randomized intervention): causality a bit easier



They predict the probability that ego is obese at time $t + 1$ based on

- ▶ alter obese at $t + 1$
- ▶ alter obese at t
- ▶ ego obese at t
- ▶ ego age
- ▶ ego sex
- ▶ ego level of education
- ▶ indicator variables for each exam period

for different types of alters

Alter Type

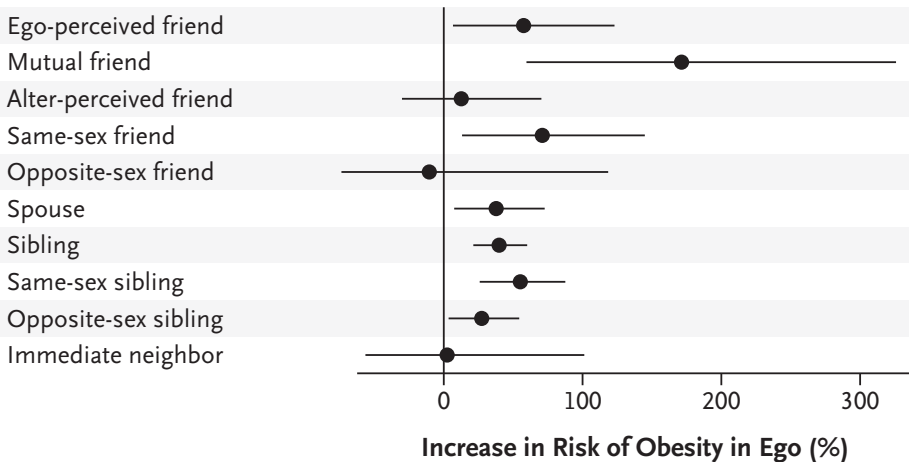
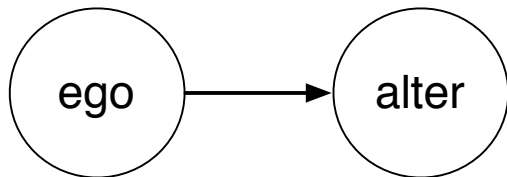
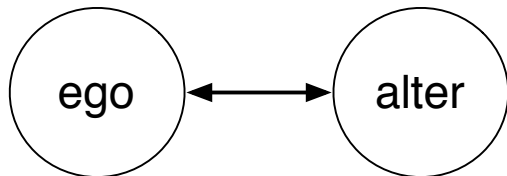


Figure 4. Probability That an Ego Will Become Obese According to the Type of Relationship with an Alter Who May Become Obese in Several Subgroups of the Social Network of the Framingham Heart Study.

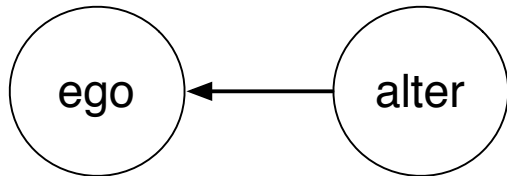
ego-perceived friend

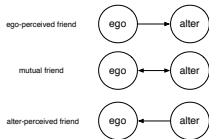


mutual friend



alter-perceived friend





If an ego stated that an alter was his or her friend, the ego's chances of becoming obese appeared to increase by 57% (95% confidence interval [CI], 6 to 123) if the alter became obese. However, the type of friendship appeared to be important. Between mutual friends, the ego's risk of obesity increased by 171% (95% CI, 59 to 326) if an alter became obese. In contrast, there was no statistically meaningful relationship when the friendship was perceived by the alter but not the ego ($P=0.70$). Thus, influence in friendship ties appeared to be directional.

Obesity in alters might influence obesity in egos by diverse psychosocial means, such as changing the ego's norms about the acceptability of being overweight, more directly influencing the ego's behaviors (e.g., affecting food consumption), or both. Other mechanisms are also possible. Unfortunately, our data do not permit a detailed examination.

**THE SPREAD OF EVIDENCE-POOR MEDICINE
VIA FLAWED SOCIAL-NETWORK ANALYSIS**

RUSSELL LYONS

C&F start by finding statistical associations between the obesity of friends in the Framingham network: To oversimplify, a person's friends are more likely to be obese if the person himself is obese. The associations that C&F analyze are calculated from statistical models whose parameters are estimated by using the observational data. This source of C&F's associations is crucial to their argument and decisive for our critique.

C&F argue that these associations are not mere associations, but measure causal effects. The two primary reasons the associations might not be causal are *homophily* (or selection), which is the fact that people tend to associate with others like themselves, and a *shared environment* (also called “confounding” or “contextual influences” by other researchers). C&F call the causal effects *induction* (also called “influence” or “endogenous social effect” by others) that they liken to a transmission process. C&F deduce induction indirectly by ruling out the possibilities of homophily and shared environment; they provide speculation, but not evidence, for how such induction might work.

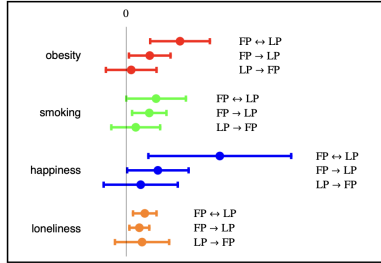


Figure 1. Coefficient estimates and 2 SE (95%) confidence intervals for directional effects. For each study, the order from top to bottom is (1) mutual friendship, (2) FP named LP, then (3) LP named FP. The CIs overlap so much that one cannot infer that the differences are statistically significant. Sources: [1, suppl. p. 3]; [2, suppl. p. 18]; [3, suppl. p. 9]; [4, pp. 983–984].

- Difference between the three is not statistically clear (true but maybe not critical)

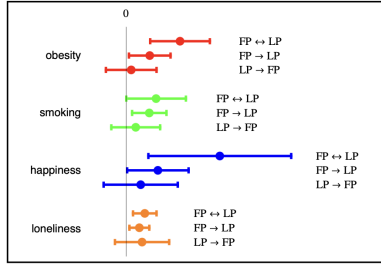


Figure 1. Coefficient estimates and 2 SE (95%) confidence intervals for directional effects. For each study, the order from top to bottom is (1) mutual friendship, (2) FP named LP, then (3) LP named FP. The CIs overlap so much that one cannot infer that the differences are statistically significant. Sources: [1, suppl. p. 3]; [2, suppl. p. 18]; [3, suppl. p. 9]; [4, pp. 983–984].

- ▶ Difference between the three is not statistically clear (true but maybe not critical)
- ▶ Even if it was clear it does not rule out homophily or common environment (true but maybe not critical)

- ▶ Christakis and Fowler have not provided mathematical proof, their data are open to other interpretations

- ▶ Christakis and Fowler have not provided mathematical proof, their data are open to other interpretations
- ▶ We should improve statistics education and peer review

- ▶ Christakis and Fowler have not provided mathematical proof, their data are open to other interpretations
- ▶ We should improve statistics education and peer review
- ▶ Christakis and Fowler and others have done lots of subsequent work. Here's how Christakis summarizes:

- ▶ Christakis and Fowler have not provided mathematical proof, their data are open to other interpretations
- ▶ We should improve statistics education and peer review
- ▶ Christakis and Fowler and others have done lots of subsequent work. Here's how Christakis summarizes:
 - ▶ “Of course, methods to estimate social contagion with observational data regarding complex networks continue to evolve, and continue to require various assumption, as ours did in 2007. As before, I would love to see someone offer a superior statistical method to observational data.”
 - ▶ “In addition, my own lab, in the span from as early as 2010 to as recently as 2024 has published many demanding randomized controlled field trials and other experiments documenting social contagion . . . ”
 - ▶ “Many other labs, in part stimulated by our work, have conducted many other experiments documenting social contagion . . . ”

Experimental studies of contagion

Does aspirin cure headaches?

pain with aspirin - pain without aspirin = effect of aspirin

But, we only get to see one outcome

Randomly give some people aspirin and some people a placebo

Compare average pain in both groups

Difference between two groups is caused by aspirin, not pre-existing differences between groups

What would that kind of experiment look like in networks?

Summary:

- ▶ “People are connected, and so their health is connected.”

Summary:

- ▶ “People are connected, and so their health is connected.”
- ▶ contagion is difficult to study in real data when people can choose their friends and may be exposed to environmental changes that we don't measure

Summary:

- ▶ “People are connected, and so their health is connected.”
- ▶ contagion is difficult to study in real data when people can choose their friends and may be exposed to environmental changes that we don't measure
- ▶ experiments provide a way forward

Next class:

- ▶ Experimental studies of contagion